

Problem

Given N floating-point numbers in the range [0, 1), sort them using the Bucket Sort algorithm.

Input Format

N a1 a2 a3 ... aN

Constraints

$1 \leq N \leq 100000$ $0 \leq a_i < 1$

Output Format

Print the sorted array rounded to 2 decimal places.

Sample Input 0

6
0.90 0.80 0.70 0.60 0.50 0.40

Sample Output 0

0.40 0.50 0.60 0.70 0.80 0.90

Leaderboard

Discussions

Sample Input 1

6
10 20 40 50 30 60

Sample Output 1

10 20 30 40 50 60

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Language Python 3



```
1 n = int(input())
2 arr = list(map(float, input().split()))
3 if max(arr) < 1:
4     buckets = [[] for _ in range(n)]
5     for num in arr:
6         index = int(num * n)
7         buckets[index].append(num)
8     result = []
9     for bucket in buckets:
10        bucket.sort()
11        result.extend(bucket)
12    for x in result:
13        print(f"{x:.2f}", end=" ")
14 else:
15    arr.sort()
16    for x in arr:
17        if x == int(x):
18            print(int(x), end=" ")
19        else:
20            print(f"{x:.2f}", end=" ")
21
```

Line: 13 Col: 35



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0



Test case 1

Compiler Message

Success